OpenAIs Astra Solves Ten Unsolved Math Problems With Verifiable Lean Proofs

On August 1, 2026, OpenAI announced a result that is already reshaping conversations across mathematics departments and AI labs alike: an internal version of its upcoming model **Astra** generated formal, machine-verifiable proofs for **ten previously unsolved mathematical problems**. Each proof was formalized in **Lean 4**, making the results checkable by a computer rather than dependent on human expert review.

The total compute cost for all ten solutions: approximately **$2,000 at API rates**.

The Problems and What Was Solved

The ten problems span several distinct fields of mathematics and theoretical computer science, and each had remained open for at least a decade. Among the highlights:

In **group theory**, Astra produced the first explicit construction of a non-sofic group — a class of mathematical object whose existence had been conjectured but never demonstrated. In **high-dimensional geometry**, it derived new asymptotic upper bounds on sphere-packing density that reach the Cohn–Elkies threshold, a theoretical ceiling that had long resisted explicit construction. In **theoretical computer science**, it proved new lower bounds for the circuit complexity of computing the permanent ($n/\log n$ formula lower bound), alongside exponential parallel repetition results for two-player quantum games.

Several of the results address problems from **Paul Erds's catalogue** of open combinatorics questions — including problems 146, 180, and 183, relating to multicolor Ramsey numbers. Other results address operator algebras (a counterexample to Connes's rigidity conjecture) and lattice problems (polynomial-factor hardness of approximation for the Closest Vector Problem).

> When a Lean formalization accompanies an AI-generated proof, the question shifts from "is the AI correct?" to "is the checker correct?" — a much more tractable problem.

Why Lean Matters Here

AI-generated mathematics has faced a persistent credibility problem: models can produce plausible-looking but subtly flawed proofs, and expert review takes time. The use of Lean 4 as a verification layer addresses this directly. Lean is a formal proof assistant that only accepts proofs that are logically complete and internally consistent. If the Lean checker accepts a proof, the mathematics is correct — not approximately correct, not probably correct.

This transforms the nature of the output. OpenAI is not publishing a set of text documents that experts need to audit; it is publishing a set of *mathematical certificates*. The 249-page collection of manuscripts and accompanying Lean formalizations are publicly available on GitHub for independent review.

Astra and Long-Horizon Reasoning

Astra is described as a model family designed for long-horizon tasks — reasoning processes that unfold over hours or days, involving coordinated multi-step planning and execution. Mathematical theorem proving is an ideal testbed for this capability: the search space is enormous, intermediate steps need to be tracked carefully, and the final answer is objectively verifiable.

The results suggest that Astra can sustain coherent multi-step reasoning across the kind of problem complexity that previously required human experts working for years. Critically, it does this in a domain with a ground-truth verification mechanism — Lean — which removes ambiguity from the evaluation.

OpenAI has made Astra available to a limited set of internal researchers. A broader release timeline has not been announced, though the August 1 publication of results signals the lab is confident enough in the findings to invite public scrutiny before a commercial launch.

For the mathematics community, the announcement marks a shift from viewing AI as a research assistant to confronting it as a potential contributor to mathematical knowledge itself — one that works faster, sleeps never, and leaves checkable receipts.

Contributor: Alessandro Linzi